A current follower has the advantage of easily obtaining large preamplifier bandwidths when using piezo transducers with values for C_{tr} of several hundred pF. The resistances R_{ph} together with capacitance C_{tr} constitute a phantom zero, which makes it possible to optimise the small-signal frequency response of the preamplifier for a given transducer. When using well known transducers R_{ph} can be integrated as well.

Experimental results: An experimental ultrasonic transceiver consisting of a class AB voltage follower as transmitter and a current amplifier as preamplifier has been integrated successfully. Fig. 3 shows a photograph of the integrated circuit, realised on a semicustom chip with a breakdown voltage of 20 V and a transit frequency $f_T(npn)$ of 300 MHz.

The specifications of the circuit are given in Table 1.

Table 1 SPECIFICATIONS OF INTEGRATED ULTRASONIC TRANSCEIVER

Maximum transmit voltage	10 V
Input impedance of transmitter	1 kΩ
Current gain of preamplifier	20 dB
Bandwidth of preamplifier	6 MHz*
Preamplifier recovery time	$<2\,\mu s$*
Total power dissipation	200 mW

* Measured with 2·25 MHz PZT transducer with $C_{tr} = 400$ pF

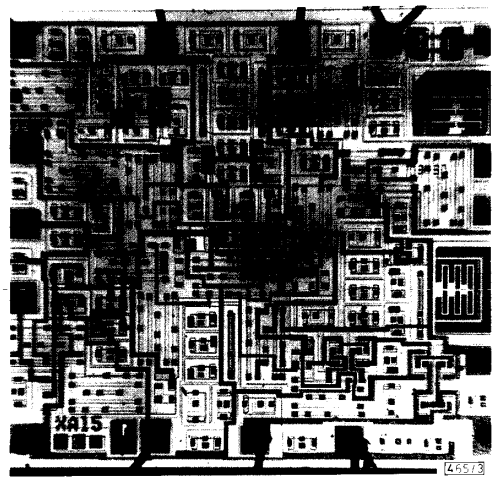

Fig. 3 *Microphotograph of integrated transceiver*

In Fig. 4 the output signal of the preamplifier is shown which results when a piezoelectric transducer with a resonance frequency of 2·25 MHz is placed above a steel reflector, both in fresh water.

The amplifier recovery time as well as the first received echo signal are visible. The transducer is fired during the first part of the amplifier recovery time with a transmit pulse of 4 V, the pulse width of the transmit pulse is approximately 200 ns.

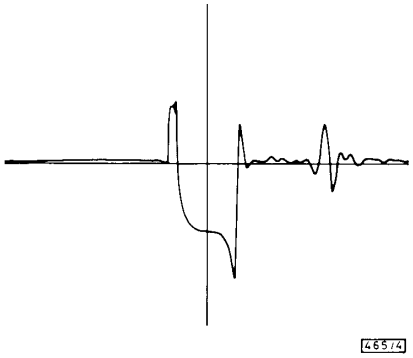

Fig. 4 *Output signal of preamplifier with 2·25 MHz piezoelectric transducer, placed above a steel reflector*

Conclusion: Single-chip ultrasonic transceivers with direct coupling between transmitter, transducer and preamplifier can be realised with circuits as described above.

The single-chip solution not only shows a remarkably short recovery time but, owing to the small size, many new applications are foreseen. The chip is presently implemented in a prototype borehole scanner for a seismic application and a prototype medical catheter for intravascular monitoring of coronair arteries.

With the presently applied IC process, the maximum transmit voltage of the transceiver is limited to 10 V.

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ANALYSIS AND DESIGN OF LLC-TYPE SERIES RESONANT CONVERTOR

Indexing terms: Circuit theory and design, Convertors

In the letter a new convertor is derived by adding an inductor in parallel with the resonant capacitor of the conventional series resonant convertor (SRC). It will be shown that the control characteristics of the new convertor are much improved when compared to those of the conventional SRC. Another advantage of the new convertor over the conventional SRC is that the component stresses are also reduced.

Introduction: At present, the parallel resonant convertor has been more widely used than the series resonant convertor (SRC) because, in the case of the SRC, the convertor gain is very much load-dependent and is difficult to control. However, the SRC has low convertor again (<1) which in some cases is an advantage for high-frequency applications. In the SRC design, considerable effort has been devoted to improving its control characteristics, using techniques such as full-bridge transistor switches, nonlinear inductor control etc.,[1,2] resulting in an increase of the circuit complicity. In this letter we present a new convertor topology of the SRC, which is derived by adding an inductor in parallel with the resonant capacitor of the conventional SRC as shown in Fig. 1. This new convertor is known as the LLC-type SRC (SRC-LLC). The capacitor C_p and inductor L_p shown in Fig. 1 introduce a zero at frequency $\omega_{op} \equiv 1/\sqrt{(L_p C_p)}$. Thus, by making ω_{0p} close to ω_0, where $\omega_0 \equiv 1/\sqrt{(LC_p)}$ and $L \equiv (L_s L_p)/(L_s + L_p)$, and operating the convertor between ω_{0p} and ω_0, the

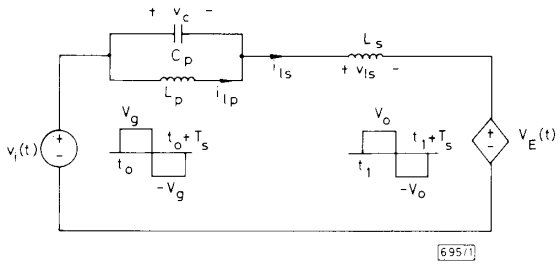

Fig. 1 *Equivalent circuit of LLC-type series resonant convertor*

maximum range of switching frequency ω_s required by different load conditions is equal to or less than $\omega_0 - \omega_{0p}$. This new convertor is known as the LLC-type SRC (SRC-LLC). By using a proper transformation of state variables, the new third-order SRC can be analysed by means of the two-dimensional state-plane diagram, similar to that of the conventional SRC. Based on the analysis results, a comparison of the new convertor and the conventional SRC is given which shows that the new convertor possesses more desirable features than the conventional SRC.

State-plane analysis: The equivalent circuit of the idealised half-bridge LLC-type series resonant convertor is shown in Fig. 1. For the continuous conduction mode of operation, the equivalent square-wave voltage source $v_i(t)$ with amplitudes $\pm V_g$ and period T_s represents the switching action of T_1/D_1 and T_2/D_2. Furthermore, we assume that C_0 is large enough so that the rectifying circuit can be replaced by a dependent voltage source $V_E(t)$ with voltage polarity depending on the sign of $i_{ls}(t)$ as illustrated in Fig. 1.

To facilitate the analysis, we define three new state variables as $v_c = v_{cp}$, $i_l = i_{ls} - i_{lp}$ and $i_p = i_{lp}$. The purpose of introducing v_c, i_l and i_p is to have v_c and i_l decoupled from i_p so that their steady-state response can be represented by a closed contour in the $v_c - i_l$ plane. In terms of the new state variables, and normalising all the voltages and currents by V_g and V_g/Z_0, respectively, where $Z_0 \equiv \sqrt{(L/C_p)}$, the normalised state-plane equation for the circuit shown in Fig. 1 is given by

$$\frac{dv_{nc}}{di_{nl}} = \frac{i_{nl}}{v_{ni}(t) - v_{nE}(t) - v_{nc}} \tag{1}$$

where

$$v_{ni}(t) \equiv \left[1 - \left(\frac{\omega_{0p}}{\omega_0}\right)^2\right]\frac{v_i(t)}{V_g}$$

$$v_{nE}(t) \equiv \left[1 - \left(\frac{\omega_{0p}}{\omega_0}\right)^2\right]\frac{v_E(t)}{V_g} \tag{2}$$

Under steady-state conditions, the solution of eqn. 1 consists of four circular arcs in the $v_{nc} - i_{nl}$ plane over a switching period T_s. The centres of these circular arcs depend on the value of $v_i(t)$ and the sign of $i_{ls}(t)$, and are located at $(V'_{ng} - V'_{n0}, 0)$, $(V'_{ng} + V'_{n0}, 0)$, $(-V'_{ng} + V'_{n0}, 0)$ and $(-V'_{ng} - V'_{n0}, 0)$, where $V'_{ng} \equiv 1 - (\omega_{0p}/\omega_0)^2$, $V'_{n0} \equiv [1 - (\omega_{0p}/\omega_0)^2]M$, and $M \equiv V_0/V_g$ which is the convertor voltage gain. Over the first half switching period when $v_i(t) = V_g$, the solution of eqn. 1 is given by

$$i_{nl}^2 + (V'_{ng} - V'_{n0} - v_{nc})^2 = i_{nl}^2(t_0) + [V'_{ng} - V'_{n0} - v_{nc}(t_0)]^2 \quad \text{for } i_{ls} > 0 \tag{3}$$

$$i_{nl}^2 + (V'_{ng} + V'_{n0} - v_{nc})^2 = i_{nl}^2(t_1) + [V'_{ng} + V'_{n0} - v_{nc}(t_1)]^2 \quad \text{for } i_{ls} < 0 \tag{4}$$

where t_0 is the initial time and t_1 is the time when $i_{ls} = 0$. Similar equations can be obtained for the second half switching period when $v_i(t) = -V_g$. A typical state-plane diagram for the steady-state solution of eqn. 1 is shown in Fig. 2.

Based on the state-plane diagram of Fig. 2, we derive the following expressions for the normalised average output and input currents:

$$I_{n0} = \frac{2}{T_s}\int_{t_0}^{t_0+T_s/2} |i_{nls}(t)|\, dt = \frac{2}{\gamma}\left[1 - \left(\frac{\omega_{0p}}{\omega_0}\right)^2\right]\left[v_{nc}(t_1) - \alpha\beta\left(\frac{\omega_{0p}}{\omega_0}\right)^2\right] \tag{5}$$

$$I_{ng} = \frac{2}{T_s}\int_{t_0}^{t_0+T_s/2} i_{nls}(t)\, dt = -\frac{2}{\gamma}\left[1 - \left(\frac{\omega_{0p}}{\omega_0}\right)^2\right]\left[v_{nc}(t_0) + \alpha\gamma M\left(\frac{\omega_{0p}}{\omega_0}\right)^2\right] \tag{6}$$

where $\gamma = \beta + \alpha = \omega_0(T_s/2) = \pi f_0/f_s$, and β and α are the conduction angles of T_1 and D_1, respectively. Since the convertor is assumed lossless, conservation of power, $I_0 V_0 = I_g V_g$, leads to the following relation:

$$M = -\frac{v_{nc}(t_0)}{v_{nc}(t_1)} \tag{7}$$

Based on eqn. 7, if V'_{ng}, M and the initial switching point $(v_{nc}(t_0), i_{nl}(t_0))$ are specified, the corresponding state-plane

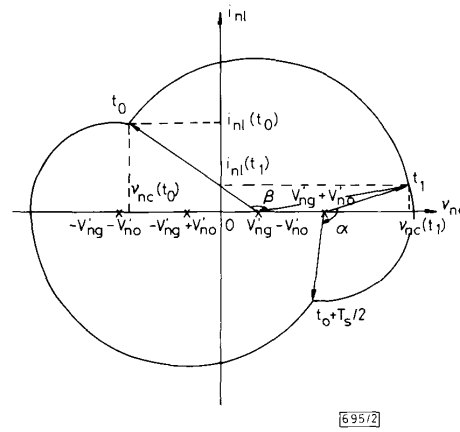

Fig. 2 *Typical state-plane diagram*

diagram can be uniquely determined. Thus, the SRC-LLC can be designed from the state-plane diagram, similar to that of the conventional SRC.[3]

Comparison between SRC-LLC and conventional SRC: Using eqns. 3 and 5–7 together, the relations between M and $f_n = f_s/f_0$ with Q_s as parameter, where $Q_s \equiv Z_0/R_0$ and R_0 is the output load, can be numerically solved, and the corresponding control curves are shown in Fig. 3 for different values of ω_{0p}/ω_0, where Fig. 3*a* represents the control curves of M

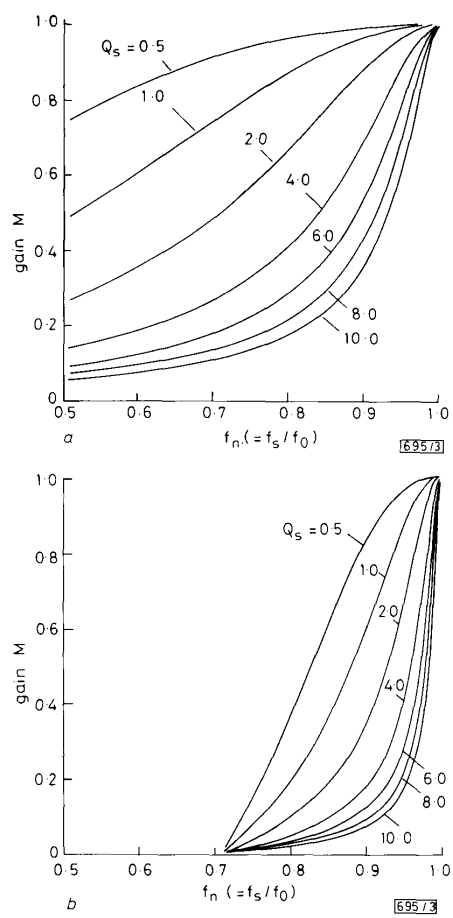

Fig. 3

a Control characteristics of conventional SRC
b Control characteristics of LLC-type SRC

against f_n for the conventional SRC ($\omega_{0p} = 0$). From Fig. 3*b* it can be seen that the control characteristics of the SRC-LLC are much improved from that of the conventional SRC, while the advantage of low convertor gain is still maintained. Fig. 4 shows the maximum normalised capacitor voltages V_{ncmax} against normalised frequency f_n with the convertor gain M as the parameter for both the conventional and LLC-type SRCs.

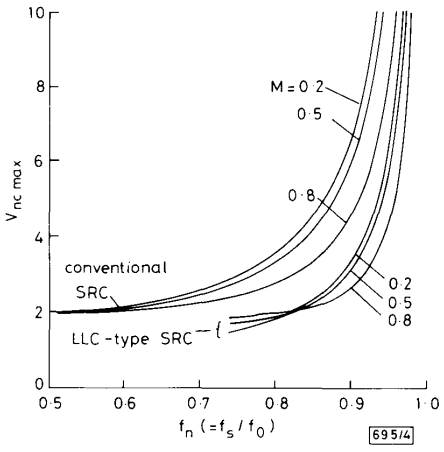

Fig. 4 *Maximum normalised capacitor voltages*

It is evident that the maximum capacitor voltage in the LLC-type SRC is reduced.

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COMPLEX RAY ANALYSIS OF REFLECTION AND TRANSMISSION OF HERMITE–GAUSSIAN BEAM AT CURVED INTERFACE

Indexing terms: EM waves, Wave propagation, Optics

The reflection and transmission of a higher-order beam field at a curved interface is investigated using the complex ray theory. A Hermite–Gaussian beam field is expressed in terms of a superposition of a finite number of complex-source-point multipole fields. By using the asymptotic approximation for the multipole fields, the complex ray analysis of a higher-order beam field is treated. The reflected and the transmitted fields of a second-order beam are calculated numerically.

Introduction: The output fields from a waveguide horn, a laser cavity and other sources emitting focusing radiation are approximated by a linear superposition of paraxial beam fields. To investigate the reflection and transmission of a Gaussian beam field at a dielectric interface is important for the design of optical devices such as lenses used in optical fibre transmission systems. For the phenomenon of the reflection and transmission of a lowest-order beam field at a dielectric interface, much work has been done[1–3] so far. As far as we know, no analysis has been presented for the reflection and transmission of a higher-order beam field at a curved interface.

In this letter we investigate the reflection and transmission of a higher-order beam field at a curved interface using the complex ray method. In a previous paper[4] we showed that a higher-order beam field is expressed in terms of complex-source-point multipole fields. We apply the asymptotic approximation to the multipole fields and derive the ray description of a higher-order beam field. As a numerical example, we calculate the reflected and transmitted fields of a second-order beam field.

Analysis by complex ray method: Fig. 1 shows the complex ray geometry of the transmitted field by a line source at a complex

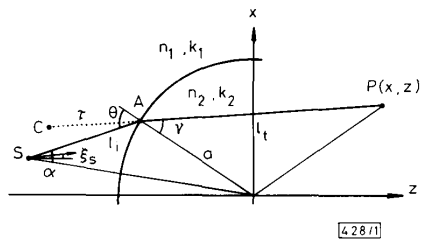

Fig. 1 *Geometry of problem*

location $S(x_s - jb_s \sin \xi_s,\ z_s - jb_s \cos \xi_s)$. In this Figure the refractive indices n_1 and n_2, the wavenumbers k_1 and k_2, the radius of curvature of the interface a, the angle ξ_s between the propagation axis of the incident beam field and the z-axis and the co-ordinate (x, z) of the observation point P are real and the other parameters are complex. We have shown that a higher-order beam field is represented in terms of a superposition of a finite number of complex-source-point multiple fields.[4] By using the asymptotic approximation for the multipole fields, the ray description of an incident nth-order beam field E_n^i at point A is given as follows:

$$E_n^i \sim (-1)^n k_1 \left(\frac{\pi}{2}\right)^{1/4} \left(\frac{n!\, w_s}{2}\right)^{1/2} e^{-k_1 b_s} \left(\frac{2j}{\pi k_1 l_i}\right)^{1/2} \times \exp(-jk_1 l_i) g_n(k_1, w_s, l_i, \psi_s) \quad (1)$$

where

$$g_n(k_1, w_s, l_i, \psi_s) = \sum_{p=0}^{[n/2]} \sum_{q=0}^{n-2p} \frac{2^{-p}}{p!\,(n-2p)!} \times \left(j\frac{k_1 w_s}{2}\right)^{n-2p} \binom{n-2p}{q} \times \exp\left[j\left\{(n-2p-2q)\left(\psi_s + \frac{\pi}{2}\right)\right\}\right] \times \sum_{r=0}^{\infty} \frac{(n-2p-2q, r)}{(2jkl_i)^r} \quad (2)$$

$$(m, r) = \frac{(4m^2-1^2)(4m^2-3^2)\cdots[4m^2-(2r-1)^2]}{2^{2r} r!}$$

$$(m, 0) = 1 \quad (3)$$

and $[n/2]$ is $n/2$ or $(n-1)/2$ according to whether n is even or odd, $\binom{n}{q}$ is the binomial coefficient, l_i is the complex distance from the source point S to point A, ψ_s is the complex angle between the propagation axis of the incident beam and the complex ray trajectory SA, and w_s $[= \sqrt{(2b_s/k_1)}]$ is the smallest spot size of the lowest-order beam field whose centre is at (x_s, z_s).

When we apply the paraxial approximation and neglect the terms with powers not less than f $(f = 1/k_1 w_s)$ in eqns. 1 and 2, the beam description of a higher-order beam field is obtained.